THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the right conditions for an organism to live and grow	Home of an organism
2	**(a)** bacteria	Living organism
3	**(b)** green plants that make their own food	First trophic level
4	**(d)** bacteria and fungi	Break down dead matter
5	**(c)** food web	More realistic picture
6	**(c)** trophic levels	Troph = food
7	**(b)** mutualism	Both gain
8	**(a)** farm	Created by humans
9	**(d)** mangrove forests	At Ganga–Brahmaputra delta
10	**(d)** compete with native species	Not native

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Habitat: place providing right conditions for an organism (1). Ecosystem: interaction of biotic and abiotic components (1).	1+1
12	Biotic: plants, microbes (1). Abiotic: air, water (1).	1+1
13	Producers (1). Consumers and decomposers (1).	1+1
14	Sequence showing who eats whom (1). Grass → grasshopper → frog → snake (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Producers make food, e.g. plants (1). Consumers eat others, e.g. deer, lion (1). Decomposers break dead matter, e.g. fungi (1).	1+1+1
16	Four organisms in order (1). Labelled producer, primary, secondary and tertiary consumers (1). Arrows showing direction of food (1).	1+1+1
17	Arrows show flow of energy and matter (1½). Energy passes from prey to predator (1½).	1½+1½
	OR: Reducing frogs raises insects (1). Fewer snakes find food (1). Balance of the food web changes (1).	1+1+1
18	Mushrooms and bacteria decompose dead matter (1). Nutrients return to the soil (1). Plants take them up again (1).	1+1+1
	OR: Mutualism: bee and flower (1). Commensalism: orchid on tree (1). Parasitism: tapeworm in humans (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Biotic defined (1). Abiotic defined (1). Pond biotic examples (1). Pond abiotic examples (1). Interactions (1).	1+1+1+1+1
	OR: Food chain (1). Food web (1). Six organisms (1). Arrows (1). Trophic levels marked (1).	1+1+1+1+1
20	Trophic levels (1). Producers at base (1). Consumers above (1). Energy decreases upward (1). Fewer organisms at top (1).	1+1+1+1+1
	OR: Decomposers feed on dead matter (1). Break complex matter into simpler (1). Compost forms (1). Nutrients return to soil (1). Cycle continues (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** biotic components	Plants, animals, microbes
2	**(c)** biotic and abiotic components	Living and non-living
3	**(d)** herbivores	Primary consumers
4	**(b)** saprotrophs	Feed on rotten matter
5	**(a)** grass	First level
6	**(d)** producers	They make food
7	**(b)** commensalism	Orchid benefits; tree unaffected
8	**(a)** monoculture	Reduces biodiversity
9	**(a)** slowing down winds and waves	Natural barrier
10	**(d)** conservation of ecosystems	Protected areas

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Network of interconnected food chains (1). Shows many feeding relationships in an ecosystem (1).	1+1
12	Positions of organisms in a food chain (1). Producers occupy the first level (1).	1+1
13	They break down dead plants and animals (1). Recycle nutrients to soil (1).	1+1
14	Both organisms benefit (1). Bee and flower (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Millet at the base (1). Mouse, snake above (1). Eagle at top, fewer in number (1).	1+1+1
16	Pollination of crops (1). Natural pest control (1). Fertile soil from decomposers (1).	1+1+1
17	Pests become resistant (1). Pollinators decrease (1). Biodiversity falls (1).	1+1+1
	OR: Organic manure (1). Crop rotation (1). Reducing synthetic chemicals (1).	1+1+1
18	Ancient text on plant care (1). Stresses soil nourishment by organic manure (1). Kunapa Jala is a fermented liquid fertiliser (1).	1+1+1
	OR: Rising sea levels (1). Pollution and overuse of resources (1). Habitat loss affects wildlife (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mutualism (1½). Commensalism (1½). Parasitism (1). Examples (1).	1½+1½+1+1
	OR: Farms, fish ponds (1). Pollination (1). Pest control (1). Soil nutrients (1). Water retention (1).	1+1+1+1+1
20	Pests resistant (1). Pollinators decline (1). Biodiversity falls (1). Organic farming (1). Crop rotation and manure (1).	1+1+1+1+1
	OR: Largest mangrove forest (1). Protects from storms (1). Biodiversity (1). World Heritage Site (1). Threats like pollution and sea-level rise (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.